

Security, Performance & Monitoring Report

Vagdevi Malineni

Nainisha

GitHub Repo: [Nandini-Bhimineni/Raritone-Project-Backend at feature/security-performance](#)

Module Name

Security, Performance & Monitoring

Branch Name

feature/security-performance

Implemented Features

1. Joi Validation

Installed Joi package for backend validation.

Validations Added

- Login validation
- Register validation
- Product upload validation
- Try-on request validation

Validation Benefits

- Prevents invalid API requests
- Improves API security

- Reduces backend crashes
- Ensures proper data formatting

2. Morgan Logging

Installed Morgan middleware.

Added Middleware

```
app.use(morgan('dev'));
```

Benefits

- Logs API requests
- Helps monitor backend activity
- Useful for debugging
- Tracks request status and response time

3. JWT Security

Verified JWT authentication middleware.

Security Features

- Protected APIs
- Token-based authentication
- User authorization
- Secure route access

4. Environment Variables

Verified secure environment variable usage.

Variables Used

- PORT
- MONGO_URI
- JWT_SECRET
- CLOUDINARY_CLOUD_NAME
- CLOUDINARY_API_KEY
- CLOUDINARY_API_SECRET

Benefits

- Prevents exposing secrets
- Improves security
- Easier environment management

5. Rate Limiting Research

Purpose

Protect APIs from abuse and excessive requests.

Recommended Package

```
npm install express-rate-limit
```

Benefits

- Prevents brute-force attacks
- Reduces server overload
- Improves backend stability

6. Redis Caching Research

Purpose

Improve backend performance using caching.

Benefits

- Faster API responses
- Reduced database load
- Better scalability
- Session caching support

7. Sentry Monitoring Research

Purpose

Real-time backend error monitoring.

Benefits

- Tracks production errors
- Crash reporting

- Faster debugging
- Performance monitoring

8. Grafana Monitoring Research

Purpose

Visualize backend metrics and logs.

Features

- Real-time dashboards
- API monitoring
- Server performance tracking
- Alert systems

9. Firebase Analytics Research

Purpose

Track application usage and analytics.

Benefits

- User activity tracking
- Event monitoring
- Performance analytics
- App engagement analysis

10. CDN Optimization Research

Purpose

Improve image and asset delivery speed.

Recommended CDN

- Cloudinary CDN
- Firebase CDN

Benefits

- Faster loading
- Reduced latency
- Better scalability
- Optimized image delivery

11. Compression Research

Recommended Package

```
npm install compression
```

Benefits

- Smaller API response sizes
- Faster response time

- Reduced bandwidth usage

Deliverables Completed

- Joi validation
- Morgan logging
- JWT verification
- Environment variable verification
- Security research
- Performance optimization research
- Monitoring tools research
- Documentation completed

Technologies Used

- Node.js
- Express.js
- Joi
- Morgan
- MongoDB
- JWT
- Cloudinary

Status

Module Completed Successfully.